

Core Features

1. Citizen-Owned Digital Identity (Self-Sovereign Identity)

Each citizen receives a secure digital identity that they can control and use across different services. The system reduces dependency on repeated manual verification.

2. Decentralized Identity (DID) and Verifiable Credentials (VC)

Government-verified documents and credentials are converted into trusted digital credentials. Organizations can verify information without asking citizens to submit the same documents repeatedly.

3. Biometric Identity Verification

The platform uses biometric verification such as face or fingerprint authentication together with government data verification to ensure that an identity belongs to the correct person.

4. Privacy-Preserving Verification with Zero-Knowledge Proofs (ZKP)

Citizens can prove required information without revealing personal details. This protects privacy while allowing secure verification.

5. Secure Citizen Digital Wallet

A wallet application allows citizens to store their digital credentials, manage identity permissions, view activity history, and receive digital rewards.

6. Blockchain-Based Trust and Audit Trail

Blockchain keeps important verification records, transactions, and system activities tamper-resistant. This creates transparency and allows authorized parties to audit activities.

7. Green Citizen Token Reward System

The platform encourages environmentally responsible behavior by rewarding verified eco-friendly actions through a token-based incentive mechanism.

8. IoT-Based Eco-Action Verification

IoT devices such as smart recycling booths can automatically verify environmental activities. Only verified actions can generate rewards, reducing fake submissions.

9. Reputation and Civic Contribution Score

Verified activities and positive contributions can build a citizen reputation score that encourages responsible behavior.

10. Credential Management and Revocation

Authorized authorities can issue, update, or revoke credentials when necessary, maintaining trust in the identity ecosystem.

11. Layered On-Chain and Off-Chain Architecture

Sensitive personal information is stored securely outside the blockchain, while blockchain stores verification proofs, transactions, and important records.

12. Government and Organization Integration

API-based integration allows connection with government databases, financial institutions, universities, and other service providers.

13. Cross-Border Identity Interoperability

The platform is designed to support future international verification of trusted credentials across different systems.